Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `test` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, direct-mapped L1D/L2 · gem5 `e0a12aa4e3+dirty`

Superseded results (metric schema 8). The current detector (schema 9) also counts a reload whose address lies inside a wider or merged store (e.g. two 4-byte spills read by one 8-byte load). Schema 8 missed those, so `spill_load`, `spill_event` and the `other_stack` counts here are lower (up to 15% of the spill reloads in bwaves, 8.5% in leela, 3% in x264, under 0.1% in most benchmarks); instruction counts, spill stores and the cache behaviour of the pairs it found are unaffected.

Definitions (metric schema 8). Every stack store issued inside the region of interest, and every later load that reads it back (a load from the same address at the same width), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	48 023 254 960	3 817 969 294	3 794 919 348	22 956 427	93 519	99.4	21.32	0.10	17.357	48.80	6.419	5.506	1.503	0.285
parest	37 262 401 281	126 952 414	16 148 158	110 058 554	745 702	12.7	0.22	0.59	15.359	47.58	20.722	6.626	4.432	5.347
deepsjeng	46 202 686 169	1 436 088 677	380 604 124	1 046 384 737	9 099 816	26.5	2.78	4.53	13.686	19.74	6.735	9.452	4.189	4.775
leela	26 063 891 644	802 403 734	20 913 175	749 189 404	32 301 155	2.6	0.34	5.75	8.956	17.75	20.160	0.455	3.522	4.806
namd	31 914 751 099	336 698 329	334 454 030	1 939 340	304 959	99.3	3.13	0.01	15.582	37.33	12.114	2.494	4.501	4.822
lbm	6 506 800 040	274 131 044	274 130 500	544	0	100.0	9.67	0.00	27.221	75.06	1.630	5.456	15.074	0.184
nab	6 691 483 051	150 911 281	124 535 928	24 348 560	2 026 793	82.5	7.90	0.73	10.786	15.30	1.540	1.648	2.584	2.927
xz	2 014 351 521	32 945 274	4 682 897	26 392 608	1 869 769	14.2	0.65	2.62	8.704	28.28	1.423	0.574	2.159	2.774
xz_4m_l1	2 001 462 938	29 215 304	3 958 947	23 548 312	1 708 045	13.6	0.57	2.35	10.079	36.30	1.566	0.593	2.726	1.808
xz_4m_l2	2 083 103 943	28 125 372	3 700 138	22 776 791	1 648 443	13.2	0.51	2.19	11.652	42.38	1.613	0.611	3.599	2.134
xz_4m_l3e	10 573 516 663	124 670 474	54 899 409	66 069 008	3 702 057	44.0	2.44	1.25	9.401	25.69	3.211	2.596	3.576	3.921
xz_4m_l4	4 011 967 585	66 558 242	20 142 751	44 411 548	2 003 943	30.3	2.11	2.21	10.814	33.53	3.105	3.152	3.148	2.674
xz_4m_l4e	12 115 711 352	127 720 342	56 348 941	67 532 844	3 838 557	44.1	2.22	1.11	9.165	24.56	3.325	2.620	3.659	4.145
xz_1m_l0	711 271 503	12 737 212	1 091 050	10 938 543	707 619	8.6	0.56	3.08	7.980	29.63	2.299	0.928	1.658	1.510
xz_1m_l1	719 848 327	12 124 762	973 253	10 458 508	693 001	8.0	0.52	2.91	8.793	33.78	2.452	0.935	1.717	1.579
xz_1m_l2	724 690 222	11 886 991	956 147	10 241 945	688 899	8.0	0.51	2.83	8.922	34.53	2.470	0.939	1.797	1.667
xz_1m_l3e	1 994 406 562	32 343 460	11 257 805	20 010 020	1 075 635	34.8	2.49	2.01	8.680	23.89	2.864	3.225	1.979	3.759
xz_1m_l4	1 076 057 785	22 448 482	5 149 009	16 486 476	812 997	22.9	1.97	3.06	8.845	28.51	2.999	4.939	1.754	3.271
xz_1m_l4e	2 198 787 666	32 786 002	11 469 277	20 222 586	1 094 139	35.0	2.33	1.84	8.884	22.40	2.964	3.234	2.007	3.876
imagick	117 733 663	2 444 391	2 450	2 437 084	4 857	0.1	0.02	4.14	7.434	47.92	16.948	70.350	13.154	0.436
blender	994 421 804	46 392 827	4 430 041	40 844 530	1 118 256	9.5	1.16	8.21	9.917	31.78	2.962	7.024	5.637	4.108
povray	2 241 026 793	168 772 970	2 121 674	155 853 333	10 797 963	1.3	0.27	13.91	13.662	37.82	13.927	21.862	3.772	4.330
omnetpp	12 426 241 547	842 378 057	5 233 032	791 024 854	46 120 171	0.6	0.09	12.73	15.066	36.61	0.724	10.669	3.331	3.571
xalancbmk	395 919 938	18 465 753	295 456	17 282 226	888 071	1.6	0.23	8.73	15.161	41.74	13.191	18.142	4.760	6.007
x264	475 535 613 470	19 188 657 460	13 318 402 468	5 322 257 634	547 997 358	69.4	6.15	2.24	3.981	26.05	1.001	1.203	1.895	2.419
gcc	16 026 870	1 020 580	52 732	952 288	15 560	5.2	0.84	11.88	10.071	32.52	3.159	4.623	3.612	2.891
mcf	24 426 982 047	234 360 896	91 410 910	142 147 076	802 910	39.0	1.58	1.16	28.220	72.37	8.955	10.166	5.465	6.451
perlbench_makerand	2 421 667 618	224 582 067	55	224 562 979	19 033	0.0	0.08	18.55	12.684	0.38	0.000	1.444	14.407	5.926
bwaves_1	17 288 358 363	145 483 579	145 451 376	26 481	5 722	100.0	3.55	0.00	44.118	27.18	5.350	1.371	4.864	6.053
bwaves_2	17 287 430 556	145 483 497	145 451 376	26 421	5 700	100.0	3.55	0.00	44.120	27.18	5.350	1.371	4.879	6.060
exchange2	57 667 143 431	1 874 700 399	1 304 759 851	345 493 349	224 447 199	69.6	5.03	1.20	4.008	1.04	3.419	3.666	4.345	3.792
fotonik3d	3 098 360 920	28 393 792	27 853 217	437 206	103 369	98.1	3.84	0.03	22.434	38.96	9.885	0.425	2.099	4.269
roms	28 955 766 090	203 000 119	182 774 988	12 238 510	7 986 621	90.0	4.79	0.08	30.012	38.90	7.109	3.376	3.203	4.921
wrf	84 396 927 626	420 494 362	109 777 107	285 403 859	25 313 396	26.1	0.91	0.68	26.370	26.50	13.692	9.129	4.905	2.812
cam4	33 466 803 465	584 634 549	168 239 124	404 426 589	11 968 836	28.8	2.11	2.42	14.018	44.30	5.423	4.410	2.154	2.157
Total	1 003 626 868 512	31 607 981 988	**20 626 590 744**	10 039 381 174	942 010 070	65.3	5.12	2.00	0.000	—	—	—	—	—